

Adham El-Sharkawy

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Education

- **Faculty of Computing and Artificial Intelligence - Cairo University** Giza, Egypt
Bachelor of Computing and Artificial Intelligence; GPA: 2.95 Sep 2024 - Present
Expected Graduation: July 2028

Technical Skills

- **Languages:** Python, C++, Java, JavaScript
- **Frontend:** Html, CSS
- **Backend:** Django, Microsoft SQL Server, SQL, SQLite
- **Concepts:** OOP, Data Structures, Algorithms, SOLID Principles, APIs, AJAX
- **Tools:** Git

Projects

- **Maktaba - Full-Stack Library Management System (Django, APIs, AJAX)** 🔗 [Link to Repo](#)
 - Built a full-stack library system, linking a custom dashboard to a Django backend via JSON APIs and AJAX.
 - Engineered the relational database schema and implemented a custom user model for email authentication.
 - Authored an automated data-seeding script to parse, transform, and migrate legacy JSON records into the database.

- **Wafarly Shokran - Full-Stack Personal Finance System (Django)** 🔗 [Link to Repo](#)
 - Built a full-stack financial tracker with a clean UI, enabling seamless CRUD operations for user transactions.
 - Developed a Django backend using ORM queries to dynamically monitor budgets and savings goals.
 - Implemented an automated SMTP email alert system to notify users when spending exceeds 80% of limits.

- **Object-Oriented Board Games Implementation (C++, OOP)** 🔗 [Link to Repo](#)
 - Implemented game logic for Ultimate Tic-Tac-Toe and similar games using a provided base engine.
 - Applied OOP principles to extend classes and manage game rules and states.

- **Image Processing Editor "Baby Photoshop" (C++, Algorithms)** 🔗 [Link to Repo](#)
 - Built a desktop app for loading and manipulating images, focusing on efficient memory management.
 - Implemented custom image processing algorithms and filters, including grayscale, blur, and edge detection.

- **Desktop Audio Player Application (C++, JUCE Framework)** 🔗 [Link to Repo](#)
 - Developed a standalone desktop audio player featuring a custom GUI built with modern C++ and JUCE.
 - Integrated advanced functionalities, including waveform display, playlist support, and metadata extraction.
 - Applied a modular Object-Oriented Architecture to ensure a scalable and highly maintainable codebase.

- **Student Management System (C++, OOP)** 🔗 [Link to Repo](#)
 - Engineered an academic administration tool to efficiently track and manage student records and grades.
 - Implemented sorting algorithms to dynamically generate and display merit-based leaderboards.

Accomplishments

- **Generative AI Training Camp:** Completed a specialized training program organized by the Information Technology Institute (ITI) in collaboration with NVIDIA, focusing on LLMs, LangChain, and RAG architectures.
- **Competitive Programming:** Active participant in Codeforces contests, focusing on Data Structures and Algorithms problems using C++. [Codeforces Profile](#)
- **EME - SOLVEX Competition 2026:** Participated in a team-based problem-solving challenge organized by ITIDA - EME Innovation Labs. Applied analytical thinking and algorithmic skills using C++ to solve complex problems under time constraints on Codeforces.